MAT1100 Homework 2

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October 12 2026

Question 5

Suppose that $|G| = p^\alpha m$ with $p \nmid m$ and $|N| = p^\beta n$ with $p \nmid n$ then it follows that $\beta \leq \alpha$ and $n \mid m$.

(a) *Proof.* $P \cap N \leq P$ so it is a p-group by Lagrange's theorem. and suppose it has order p^a with $a \leq \beta$

Now N is normal so $PN \leq G$ and

$$|PN| = \frac{|P||N|}{|P \cap N|} = p^{\alpha+\beta-a}n$$

now $|PN| \mid |G| = p^\alpha m$ by Lagrange's theorem and $\gcd(p, m) = 1$ and p is prime which implies that

$$p^{\alpha+\beta-a} \mid p^\alpha$$

and hence $\beta = a$. Hence $P \cap N$ is a Sylow p-subgroup of N. □

(b) *Proof.* Suppose Q is a Sylow p-subgroup of G/N. $|G/N| = |G|/|N| = p^{\alpha-\beta}(\frac{m}{n})$ and so

$$|\pi^{-1}(Q)| = |Q||N| = p^{\alpha-\beta}p^\beta n = p^\alpha n$$

therefore a Sylow p-subgroup P of $\pi^{-1}(Q)$ is a Sylow-subgroup of G. PN is a subgroup of G with order

$$|PN| = \frac{|P||N|}{|P \cap N|} = \frac{p^{\alpha+\beta}n}{p^\beta} = p^\alpha n$$

where $|P \cap N| = p^\beta$ as it is a Sylow p-subgroup of N by (a).

Clearly

$$\pi(P) = \frac{PN}{N} \supseteq Q$$

and moreover

$$|\pi(P)| = \frac{|PN|}{|N|} = p^{\alpha-\beta}n = |Q|$$

therefore we have equality above. Now observe that since N is normal $N_G(P) \leq N_G(PN)$. This follows, as if $g \in N_G(P)$

$$gPNg^{-1} = gPg^{-1}gNg^{-1} = PN$$

therefore

$$[N_G(PN) : N_G(P)] \in \mathbb{N}$$

now $[g] \in N_{G/N}(Q)$ if and only if

$$[g]Q[g^{-1}] = Q$$

but pulling back by π this is equivalent to

$$gPNg^{-1} = PN$$

i.e.

$$\pi^{-1}\Big(N_{G/N}(Q)\Big) = N_G(PN)$$

and therefore

$$|N_{G/N}(Q)| = [N_G(PN) : N]$$

putting this together

$$\begin{aligned} n_p(G/N) &= [G/N : N_{G/N}(Q)] \\ &= \frac{[G : N]}{[N_G(PN) : N]} \\ &= [G : N_G(PN)] \\ &= \frac{[G : N_G(P)]}{[N_G(PN) : N_G(P)]} \\ &= \frac{n_p(G)}{[N_G(PN) : N_G(P)]} \end{aligned}$$

and hence $n_{G/N}(Q) \mid n_G(P)$ as $[N_G(PN) : N_G(P)] \in \mathbb{N}$. □

(c) *Proof.* Suppose $|G| = 72$ and $|N| = 9$. Then since $72 = 8 \cdot 9$ Sylow's theorem implies that there exists $H \leq G$ of order 8 and let n_8 be the number of such subgroups.

Moreover by Lagrange's theorem $N \cap H$ as their orders are relatively prime. Thus

$$|NH| = \frac{|N||H|}{|N \cap H|} = \frac{9 \cdot 8}{1} = 72$$

and therefore $NH = G$. Moreover the second isomorphism theorem implies that

$$G/N = NH/N \cong H/H \cap N = H/\{e\} \cong H$$

now by Sylow's theorem

$$n_8 = [G : N_G(H)]$$

and moreover we also have that $N_N(H) = N_G(H) \cap N$. Consider $N \curvearrowright \mathrm{Syl}_8(G)$ via conjugation. To show any two order 8 subgroups of G are conjugate by an element of N, it suffices to show that the above action is transitive. Now

$$|\,\mathrm{Syl}_8(G)| = n_8 = [G : N_G(H)]$$

and the size of the orbit of H under the above action is

$$|\mathcal{O}_H| = [N : N_N(H)] = [N : N_G(H) \cap N]$$

now since $G = HN$ and $H \leq N_G(H)$

$$N_G(H)N \geq HN$$

and hence $N_G(H)N = G$ so

$$|G| = |N_G(H)N| = \frac{|N_G(H)||N|}{|N_G(H) \cap N|}$$

and hence

$$|\operatorname{Syl}_8(G)| = \frac{|G|}{|N_G(H)|} = \frac{|N|}{|N_G(H) \cap N|} = |\mathcal{O}_H|$$

so the above action is transitive. Therefore any two subgroups of order 8 are conjugate by an element in N. $\square$